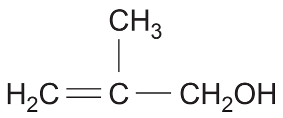
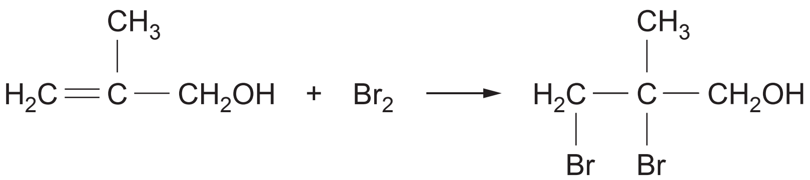
Examiner only

12. (a) 2-Methylpropenoic acid can be obtained from 2-methylprop-2-en-1-ol.



The standard reagents used for the oxidation of a primary alcohol to a carboxylic acid may affect the C═C bond. To prevent this occurring the double bond is protected by bromination.

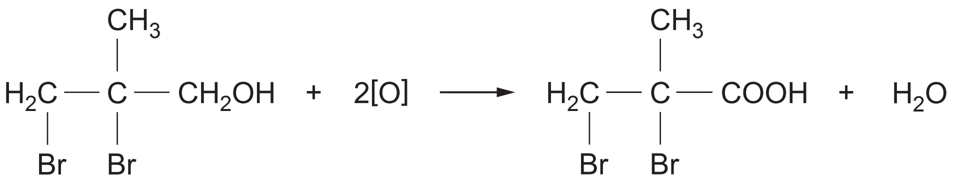


The bromine atoms are removed in a later reaction to give the required acid.

(i) State the type of mechanism occurring during the bromination. [1]

....

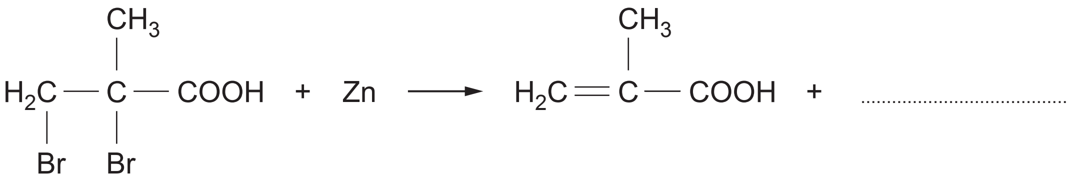
(ii) Suggest an oxidising agent used for the oxidation of the brominated alcohol. [1]



....

(iii) The dibromoacid produced in part (ii) is then reacted with zinc under suitable conditions to give 2-methylpropenoic acid.

Complete the equation, showing zinc bromide as the co-product. [1]



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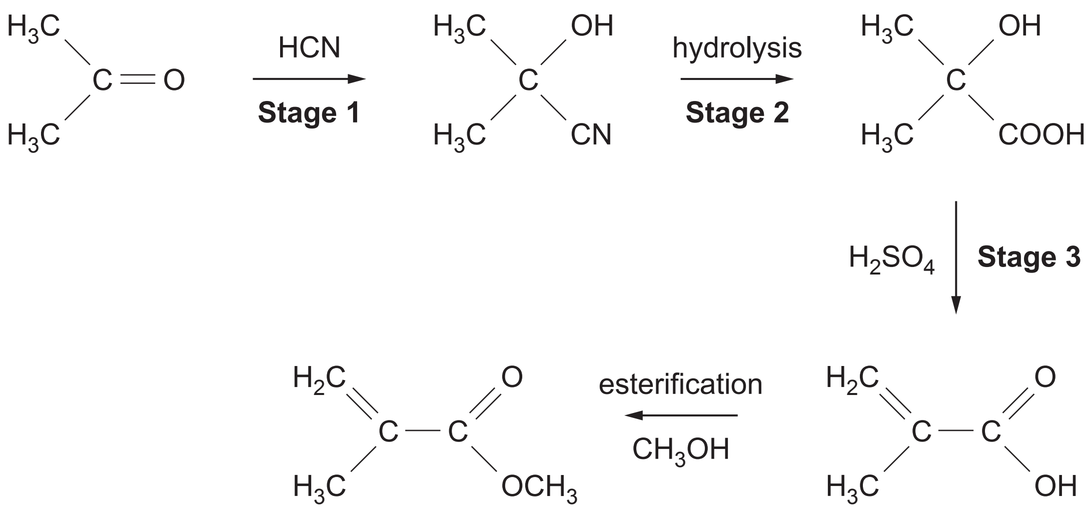
(b) Methyl 2-methylpropenoate can be produced from propanone.



(i) **Stage 1** is a nucleophilic addition reaction.

Give the formula of the nucleophile taking part in this stage. [1]

....

(ii) State why **Stage 1** is described as an **addition** reaction. [1]

....

(iii) State a reagent used for hydrolysis in **Stage 2**. [1]

....

(iv) State the role of sulfuric acid in **Stage 3**. [1]

....

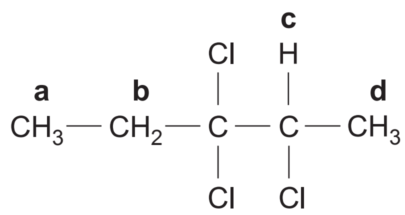
(v) Addition polymerisation of methyl 2-methylpropenoate gives ‘Perspex’.

Give the repeating unit of this polymer. [1]

Examiner only

(vi) Explain why the polymerisation in part (v) is described as addition polymerisation and not condensation polymerisation. [1]

(c) The formula of a halogenoalkane is shown below.



(i) Give the systematic name of this halogenoalkane. [1]

(ii) Complete the table below which describes the high resolution ^1H NMR spectrum of this compound. [2]

Hydrogen proton	Splitting pattern
a	
b	
c	
d	

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END OF PAPER